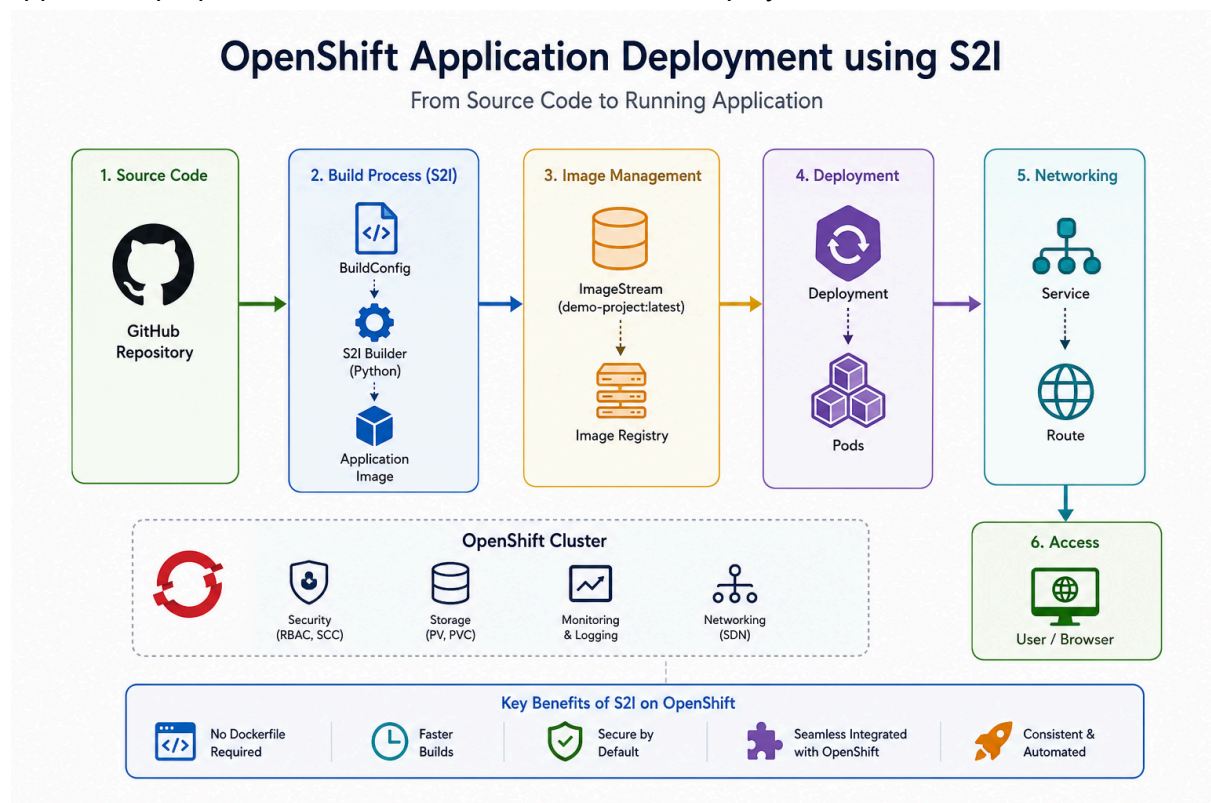


OpenShift Application Deployment using Source-to-Image (S2I)

Objective

Deploy a Python Flask application from a GitHub repository to OpenShift using the **Source-to-Image (S2I)** build process. This document covers the complete workflow from application preparation to a successful S2I build and deployment.



1. Understanding Source-to-Image (S2I)

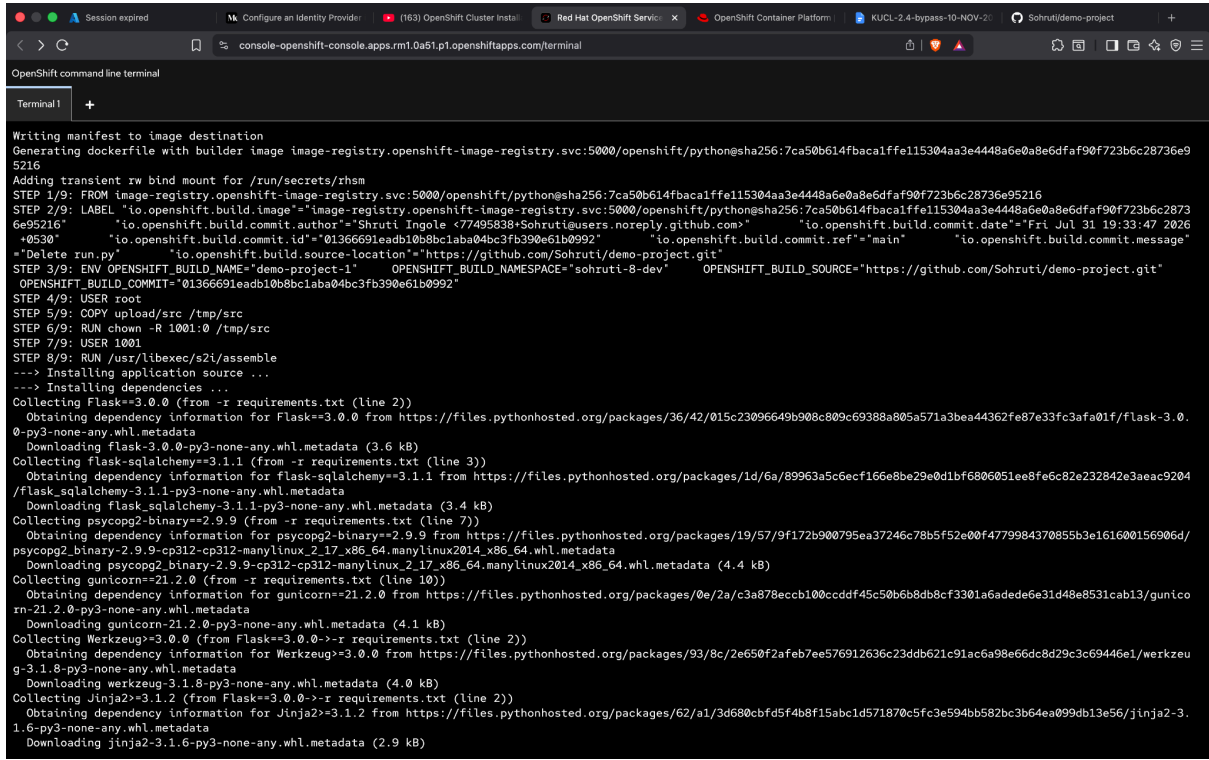
Source-to-Image (S2I) is an OpenShift feature that builds a runnable container image directly from application source code.

Unlike traditional container workflows, **no Dockerfile is required**.

2. Creating the Application

Deploy the application directly from GitHub.

oc new-app <https://github.com/Sohruti/demo-project.git>



Screenshot Placeholder: Terminal Output

3. Verifying Resources

Verify the resources created by OpenShift.

```
bash-5.1 ~ $ oc get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-project-1-build                0/1     Completed 0           2m13s
demo-project-6479b88d47-zw9jp      1/1     Running   0           67s
workspacecf81be6191477c-5fd45b8989-bvckh 2/2     Running   0           18m
bash-5.1 ~ $ oc get all
Warning: apps.openshift.io/v1 DeploymentConfig is deprecated in v4.14+, unavailable in v4.10000+
Warning: kubevirt.io/v1 VirtualMachineInstancePresets is now deprecated and will be removed in v2.
NAME                                READY   STATUS    RESTARTS   AGE
pod/demo-project-1-build            0/1     Completed 0           2m39s
pod/demo-project-6479b88d47-zw9jp  1/1     Running   0           93s
pod/workspacecf81be6191477c-5fd45b8989-bvckh 2/2     Running   0           19m

NAME                                TYPE               CLUSTER-IP      EXTERNAL-IP      PORT(S)                                     AGE
service/demo-project                ClusterIP           172.30.6.70     <none>            8080/TCP                                   2m39s
service/modelmesh-serving           ClusterIP           None            <none>            8033/TCP, 8008/TCP, 8443/TCP, 2112/TCP    7h1m
service/workspacecf81be6191477c-service ClusterIP           172.30.93.104   <none>            4444/TCP                                   19m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/demo-project        1/1     1             1           2m39s
deployment.apps/nginx               0/0     0             0           6h37m
deployment.apps/workspacecf81be6191477c 1/1     1             1           19m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/demo-project-5fd6b896f5 0         0         0       2m39s
replicaset.apps/demo-project-6479b88d47 1         1         1       93s
replicaset.apps/nginx-7cccd94f7         0         0         0       6h37m
replicaset.apps/workspacecf81be6191477c-5fd45b8989 1         1         1       19m

NAME                                TYPE      FROM      LATEST
buildconfig.build.openshift.io/demo-project Source     Git0136669 1

NAME                                TYPE      FROM      STATUS      STARTED      DURATION
build.build.openshift.io/demo-project-1 Source     Git0136669 Complete    2 minutes ago 1m8s

NAME                                IMAGE REPOSITORY                                     TAGS      UPDATED
imagestream.image.openshift.io/demo-project default-route-openshift-image-registry.apps.rm1.0a51.p1.openshiftapps.com/sohruti-8-dev/demo-project latest    About a minute ago
Error from server (Forbidden): applications.app.k8s.io is forbidden: User "sohruti-8" cannot list resource "applications" in API group "app.k8s.io" in the namespace "sohruti-8-dev"
bash-5.1 ~ $ oc expose service/demo-project
route.route.openshift.io/demo-project exposed
```

4. Exposing the Application

Expose the service using an OpenShift Route.

oc expose service/demo-project

```
bash-5.1 ~ $ oc get svc
NAME                                TYPE               CLUSTER-IP      EXTERNAL-IP      PORT(S)                                     AGE
demo-project                        ClusterIP           172.30.6.70     <none>            8080/TCP                                   4m54s
modelmesh-serving                   ClusterIP           None            <none>            8033/TCP, 8008/TCP, 8443/TCP, 2112/TCP    7h3m
workspacecf81be6191477c-service     ClusterIP           172.30.93.104   <none>            4444/TCP                                   21m
bash-5.1 ~ $ oc expose svc demo-project
Error from server (AlreadyExists): routes.route.openshift.io "demo-project" already exists
bash-5.1 ~ $ oc get route
NAME                                HOST/PORT          PATH      SERVICES      PORT      TERMINATION   WILDCARD
demo-project                    demo-project-sohruti-8-dev.apps.rm1.0a51.p1.openshiftapps.com  demo-project  8080-tcp      None
bash-5.1 ~ $
```